

Sprint 6: Purchase Reservation and Settlement

Turning a quote and a Sendaza fiat lock into safely allocated crypto backing

In progress from 2 September 2026

Current status: Sprint started. The first prerequisite now preserves existing reservations during treasury refresh. ADR-011 and the Purchase Engine contract are proposed. Purchase schema, API, settlement worker, concurrency tests, migration, and final demonstration remain pending.

Connection to Earlier Sprints

Sprint 4 supplies an exact immutable quote. Sprint 5 supplies fresh network-scoped sellable inventory. Sprint 6 will consume both only after Sendaza has created a journal-backed fiat lock. No customer balance is stored or changed inside SLE.

Proposed Flow

```
Sendaza locks exact quoted fiat
    |
    v
unexpired quote + fresh matched asset-network inventory
    |
    v
atomic crypto reservation + durable event
    |
    v
Sendaza COMMITTED -----> allocated customer backing
Sendaza ROLLED_BACK ----> reservation released
unknown / timeout -----> reconciliation; inventory held
```

Delivered So Far

Treasury synchronization now locks current inventory, preserves active reservations, and recalculates sellable units inside the snapshot transaction. This prevents a background refresh from restoring inventory that a concurrent purchase already reserved.

Pending Evidence

Real PostgreSQL oversell races, duplicate commands, quote expiry, settlement conflicts, restart/timeout behavior, immutable terminal states, outbox atomicity, API contracts, and migration verification are not yet delivered.

Glossary

Reservation: crypto inventory held for one purchase. **Allocation:** treasury backing committed to a customer crypto liability. **Fiat lock:** Sendaza's journal-backed hold on customer fiat. **Ambiguous:** an outcome that cannot safely be treated as success or failure. **Compensating record:** a new linked correction instead of editing history.

Sprint 6 remains in progress. Planned behavior must not be described as operational until its tests and release gate pass.